EDISON MENA

Senior Ignition Developer | SCADA & IIoT Solutions Engineer

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PROFESSIONAL PROFILE

Ignition Gold–Certified engineer and engineering team leader with over four years driving industrial digital transformation across data centres, mission-critical infrastructure, oil & gas, and agriculture. Designed and deployed multi-tenant Ignition platforms at scale — 500+ IIoT devices and 100,000+ tags across four countries — combining deep SCADA architecture with custom Perspective UIs, React-based components, and bespoke ESP32 firmware. Equally fluent in ISA-95 / Purdue architecture, MQTT, industrial protocols, OT cybersecurity, and modern DevOps practices. Passionate about helping integrators deliver world-class, modern industrial solutions that genuinely move the needle for their customers.

CORE TECHNICAL SKILLS

SCADA & HMI	Ignition by Inductive Automation (Gold Certified) Perspective, Vision, Tag Historian, Alarm Notification, MQTT Engine; custom Ignition SDK module development
Industrial Protocols	Modbus RTU/TCP, BACnet, SNMP, OPC UA, MQTT
Architecture & Standards	ISA-95 / Purdue Model, multi-tenant SCADA, redundant gateway design, server hardening, OT cybersecurity best practices
Programming	Jython, Python, JavaScript / TypeScript, C with FreeRTOS
Frontend & UX	React, UI/UX design, custom Perspective components, 3D rendering (GLB)
IIoT & Embedded	ESP32, ESP-IDF, FreeRTOS, custom firmware, OTA distributed updates, binary payload optimisation
Databases	MySQL, PostgreSQL
Cloud & DevOps	AWS, Docker, Git / GitHub, GitHub Actions CI/CD
Networking	CCNA-level routing, switching and security; DNS / DHCP / IPAM fundamentals
Languages	Spanish (native), English (TOEFL iBT 100/120)

PROFESSIONAL EXPERIENCE

Engineering Leader | **Xponential Technologies —** Ecuador *Jun 2024 – Present*

- Lead a multidisciplinary product team delivering **X-Monitoring**, a multi-tenant Ignition-based remote IIoT monitoring platform serving **500+ devices across four countries (Ecuador, Colombia, Bolivia, México)** with end-to-end MQTT integration.
- Architected the platform's MQTT data layer and Ignition tag model to scale beyond **100,000 tags** while preserving sub-second update performance for critical-asset monitoring.
- Designed a custom binary payload protocol that **reduced cellular data consumption per device by 90% (from ~20 MB to under 2 MB per month)**, directly cutting recurring connectivity costs across the entire fleet without sacrificing real-time fidelity.
- Engineered a fully automated **Over-The-Air (OTA) firmware update pipeline integrated with Ignition**, enabling controlled mass updates across the 500+ device fleet without site visits.
- Built and led an internal Ignition certification program: trained **five engineers to Inductive University Core and Gold level** and authored an additional in-house "Xponential Ignition Certified" curriculum to standardize delivery quality across projects.

- *Industries served: data centres and mission-critical infrastructure (UPS, HVAC, generators), floriculture, and shrimp aquaculture.*

Senior Developer Engineer | **Xponential Technologies** — Ecuador *Mar 2022 – Jun 2024*

- Designed and developed **custom firmware in C with FreeRTOS** for proprietary energy and power monitoring hardware (UPS, HVAC, generators), bridging field equipment to Ignition via standard industrial protocols.
- Co-developed a real-time **shrimp quality and traceability solution in Ignition Perspective**, integrating live GPS data from multiple IIoT devices to monitor product condition and location during transport from farm to processing plant.
- Built a **custom Ignition SDK module for Perspective enabling 3D rendering and interaction with GLB model files** — extending Perspective beyond stock components to deliver immersive, modern operator experiences.
- Contributed across the full industrial stack: embedded firmware → industrial protocols → Ignition gateway → Perspective UI → cloud reporting.

SELECTED PROJECTS

X-Monitoring — Multi-Tenant IIoT SCADA Platform — *Lead Engineer*

Multi-country IIoT platform on Ignition serving 500+ devices and 100,000+ tags across four Latin American markets. Owned the MQTT architecture, multi-tenant Ignition design, custom binary payload protocol (90% data-cost reduction), and the OTA firmware pipeline. Improved customer uptime through real-time critical-asset monitoring and early-warning alarming on UPS, HVAC and generator equipment.

Data Centre SCADA — Major Ecuadorian Bank — *Integration Engineer*

Delivered SCADA integration for the data centres of Ecuador's largest bank, covering geographically distributed sites and multiple industrial protocols. Designed the architecture to comply with the ISA-95 / Purdue model and applied server hardening and OT cybersecurity best practices for an enterprise-grade SCADA environment.

PECS — Oil & Gas Residual Treatment Plant — *Integration Engineer*

Integrated vibration data and historical process data into a centralized Ignition-based dashboard for an oil & gas residual treatment plant, giving operations a single source of truth for plant performance and condition-based monitoring.

EDUCATION

Bachelor of Engineering — Electrical, Automation and Control Engineering

Universidad de las Fuerzas Armadas (ESPE), Ecuador • *Graduated 2022*

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Ignition Gold Certified** — Inductive Automation
- **Cisco Certified Network Associate (CCNA)** — Cisco
- **AWS Certified Cloud Practitioner** — Amazon Web Services
- **Full Stack Open** — University of Helsinki (modern web development with React)
- **Foundations of Project Management** — Google

REFERENCES

Carlos Venegas
Technical Solutions Architect LATAM - CISCO
Linkedin

Eduardo Tayupantata
AI Cloud Principal - Kushki
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